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Two-sample testing up to a rotation: calibration, a projected non-Gaussianity test, and a Hermite view of contrast choice

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ARTICLE HISTORY
Compiled September 28, 2026

ABSTRACT
We test whether two whitened planar distributions are equal up to an unknown rotation—the alignment question behind two-dimensional independent component analysis. Our first message is a caution: aligning a two-sample statistic by a rotation search and calibrating by permutation is anticonservative, rejecting a genuine rotation far too often. We give a projected non-Gaussianity statistic with its limit law and a consistent multiplier bootstrap, and prove naive resampling miscalibrates. Among valid competitors an ICA-canonicalized energy test is competitive or more powerful, so the projected test's value is guaranteed calibration, not power. On real breast-cancer data the projected test is the only competitor that stays level-valid, while naive energy and even energy-after-ICA over-reject. A Hermite view explains contrast choice: whitening annihilates the first three Hermite orders, so a small corrected bank is used. All results are for $d = 2$.

KEYWORDS
rotation alignment; independent component analysis; projection pursuit; two-sample testing; multiplier bootstrap; Hermite polynomials

1. Introduction

Many questions ask not whether two distributions are equal, but whether they are equal up to a transformation one does not know. Two-dimensional independent component analysis (ICA) asks whether two whitened planar laws agree up to a rotation. This paper is about testing that alignment, and about a pitfall in doing so.

1.0.0.1. A calibration pitfall.. The obvious recipe is to take a good two-sample statistic, minimise it over rotations of one sample, and calibrate by permutation. This is invalid. The rotation search biases the observed statistic downward in a way the permutation replicates only partially, and the resulting test rejects a true rotation far too often (Section 5; over 40% in our runs). Testing equality up to a group action is not a cosmetic modification of testing equality: the nuisance must be handled in the calibration.

1.0.0.2. When the question is nontrivial.. Whitening removes second-order structure. If the two laws have anisotropic covariance, whitening aligns them to their

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covariance eigenaxes and the rotation is already resolved; a plain two-sample test then suffices. The alignment question is substantive only in the isotropic regime, where covariance does not pin the rotation and only higher-order structure can. We work there throughout.

1.0.0.3. What is actually tested.. Our statistic compares, after the best alignment, the direction *profiles* of a scalar contrast. A profile is a projection summary and does not determine the law, so profile alignment is *implied by* but does not *imply* the alignment null. The test is thus a calibrated screening test for a profile-based surrogate; a small bank of contrasts widens the surrogate but does not make it exact. We state this plainly rather than claim a characterisation the statistic cannot deliver.

1.0.0.4. Contributions.. (i) A limit law for the aligned statistic with a consistent multiplier-bootstrap calibration, and a proof that naive resampling is anticonservative (Sections 3–4); (ii) a comparison that sorts competitors into valid and invalid, honest that among the valid tests an ICA-canonicalized energy test is competitive or better (Section 5); (iii) a Hermite analysis of contrast choice (Section 6). All results are for $d = 2$; Section 8 discusses general d.

1.0.0.5. Related work.. Two-sample equality is tested by the energy distance [1], kernel MMD [2], and, for dependence, the copula tests of Rémillard and Scaillet [3]; none targets equality up to a group action. Projection indices of non-Gaussianity go back to projection pursuit [4, 5] and underlie ICA [6, 7]; identifiability of the rotation is the ICA identifiability question [6], with modern estimation theory in [8, 9]. Our focus is a calibrated test of the rotation's existence, not its estimation.

2. Setup

Let X, Y be $\mathbb{R}^2$-valued with finite moments and identity covariance ("whitened"; in the isotropic regime whitening is a standardisation that leaves the rotation unresolved). We observe independent i.i.d. unpaired samples of size n.

Definition 1 (Alignment null)**.** *With $\mathcal{O}(2)$ the orthogonal group, H_0 : $\mathrm{law}(Y) = R_{\#}\mathrm{law}(X)$ for some $R \in \mathcal{O}(2)$ ($R_{\#}$ = pushforward).*

Fix a contrast $\xi : \mathbb{R} \to \mathbb{R}$. For $a_\theta = (\cos\theta, \sin\theta)$ and a whitened law P, the *profile* is $k_P(\theta) = \mathbb{E}_{Z\sim P}[\xi(a_\theta^\top Z)]$. A rotation acts on the index by a shift, a reflection by $\theta \mapsto -\theta$, so under H_0 the profiles agree after a shift. Define the profile discrepancy

$$D_\xi = \min_{\epsilon\in\{\pm 1\}} \min_{s} \int_0^{2\pi} \big(k_Y(\theta) - k_X(\epsilon\theta + s)\big)^2 \, d\theta.$$

Remark 1 (Surrogate hypothesis)**.** D_ξ tests H_0^ξ : "the ξ-profiles align", not H_0 itself. Since $H_0 \Rightarrow H_0^\xi$ but not conversely, the test controls the level under H_0 and is a screening test for it: it can miss alternatives whose profiles happen to align (Section 6 makes this precise in Hermite terms). This is the price of reducing a 2-D comparison to a 1-D profile; the ICA-canonicalized energy test of Section 5, which retains the full distribution, does not pay it, at the cost of a canonicalisation heuristic.

With empirical profiles $\hat{k}_X, \hat{k}_Y$ on a fixed grid $\theta_1, \dots, \theta_M$, the statistic is

$$T_n = n \cdot \min_{\epsilon \in \{\pm 1\}} \min_{s} \frac{1}{M} \sum_{g=1}^{M} \big(\hat{k}_Y(\theta_g) - \hat{k}_X(\epsilon\theta_g + s)\big)^2. \tag{1}$$

3. Limit law

Assumption 1 (ξ-integrability and Donsker class)**.** *ξ measurable,* $\sup_a \mathbb{E}[\xi(a^\top Z)^2] < \infty$ *for both laws, and* $\{z \mapsto \xi(a^\top z) : a \in \mathbb{S}^1\}$ *Donsker for both.*

Lemma 1 (Functional CLT)**.** *Under Assumption 1,* $\sqrt{n}(\hat{k}_X - k_X) \rightsquigarrow \mathbb{W}_X$ *and* $\sqrt{n}(\hat{k}_Y - k_Y) \rightsquigarrow \mathbb{W}_Y$ *independently, mean-zero Gaussian with covariance* $\Gamma_\xi^\bullet(\theta, \theta') = \mathrm{Cov}(\xi(a_\theta^\top Z), \xi(a_{\theta'}^\top Z))$.

Let s_0 be the population-optimal shift ($D(s_0) = 0$ under H_0).

Assumption 2 (Non-degenerate alignment)**.** *s_0 is isolated modulo reflection,* $D''(s_0) > 0$, $k_X'(\cdot + s_0) \not\equiv 0$.

Theorem 1 (Limit law)**.** *Under H_0 and Assumptions 1–2, for a fixed grid of M directions,*

$$T_n \rightsquigarrow L_\xi^{(M)} := \|\Delta\|^2 - \frac{\langle \Delta, k_X'(\cdot + s_0)\rangle^2}{\|k_X'(\cdot + s_0)\|^2}, \quad \Delta = \mathbb{W}_Y - \mathbb{W}_X(\cdot + s_0),$$

a finite weighted sum of independent χ_1^2 variables (weights = eigenvalues of the covariance of Δ off the score direction). As $M \to \infty$ the discrete inner products converge to their integral analogues.

Remark 2 (Fixed grid)**.** We treat M as fixed; T_n converges to the M-dimensional quadratic form $L_\xi^{(M)}$, not to a continuous functional. This is deliberate: the bootstrap of Section 4 uses the *same* grid, so it reproduces $L_\xi^{(M)}$ for that M and validity does not require $M \to \infty$. Larger M refines the alignment resolution at $O(M)$ cost.

Remark 3 (Role of the limit law)**.** $L_\xi^{(M)}$ is not a plug-in critical value—it depends on unknown moments. It is the law the bootstrap reproduces, and under contiguous alternatives $k_Y = k_X(\cdot + s_0) + h/\sqrt{n}$ it becomes the noncentral $\|\Delta + \Pi h\|^2$ (Π = projection off the score), giving the local power.

4. Calibration and consistency

4.0.0.1. Why naive calibration fails.. A permutation two-sample bootstrap tests equality of laws, not equality up to rotation, and rejects true alignments; a plug-in dropping the score term overstates the statistic; a split estimating the alignment separately mis-centres the correction. Most consequentially (Section 5), *minimising a two-sample statistic over rotations and permuting* is anticonservative: the rotation search lowers the observed value more than it lowers the permutation replicates.

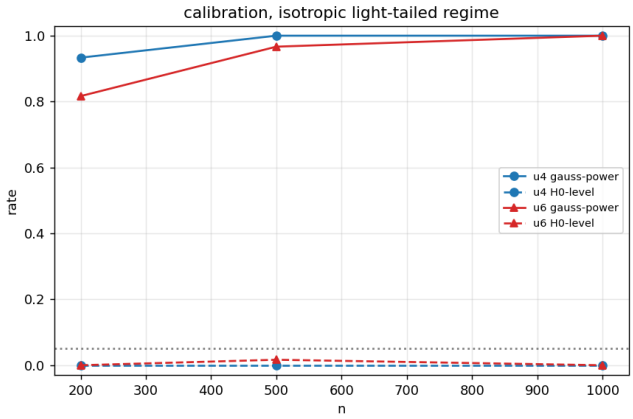


Figure 1. Calibration, isotropic light-tailed regime. Dashed: null level; solid: power vs a Gaussian alternative; dotted: nominal 0.05.

4.0.0.2. Multiplier bootstrap.. Let $\hat{\psi}_g^X, \hat{\psi}_g^Y$ be the whitened profile influence functions and $w_1, \dots, w_n$ (resp. $w_1', \dots$) i.i.d. mean-zero variance-one multipliers independent of the data. With $\mathbb{W}_X^*(\theta_g) = n^{-1/2} \sum_i w_i \hat{\psi}_g^X(X_i)$ and $\mathbb{W}_Y^*$ analogously, form T_n^* from $\Delta^* = \mathbb{W}_Y^* - \mathbb{W}_X^*(\cdot + \hat{s})$ exactly as T_n (the same aligned minimisation reproduces the score correction).

Proposition 1 (Bootstrap consistency)**.** *Under H_0 and Assumptions 1–2, conditionally on the data and in probability $T_n^* \rightsquigarrow L_\xi^{(M)}$; the test rejecting when T_n exceeds the conditional $(1-\alpha)$ quantile has asymptotic level α.*

Proposition 2 (Estimated whitening)**.** *With sample whitening, Lemma 1 holds with ψ_g^X replaced by $\bar{\psi}_g^X = \psi_g^X + \Lambda_g^\top \operatorname{vech}(XX^\top - \mathbb{E}XX^\top)$; the bootstrap stays consistent if the multipliers weight $\bar{\psi}$.*

Proposition 3 (Consistency)**.** *If $D_\xi > 0$ then $T_n \to \infty$ and the power tends to one. If $D_\xi = 0$ while H_0 fails (Remark 1), the test is not consistent for that contrast—the reason for the bank of Section 6.*

4.0.0.3. Numerical calibration.. Figure 1 shows the bootstrap in the isotropic light-tailed regime: the level is controlled—indeed conservative at these sizes—and power tends to one.

5. Which tests are valid, and how they compare

We compare five procedures at $n = 200$ over 80 replications (Figure 2, Table 1): the projected test; an ICA-canonicalized energy test (canonicalise each sample by FastICA, then an energy two-sample test); a rotation-aligned energy test (minimise energy over rotations, permutation-calibrated); a naive energy test; and a naive kernel (MMD) test.

Two findings stand out. First, *validity*: aligning a two-sample statistic by a rotation search and permuting does not control the level—the rotation-aligned energy test rejects the rotation null 54% of the time, and the naive energy test 84%. Only the projected test and energy-after-ICA are valid. Second, *power*: among the valid

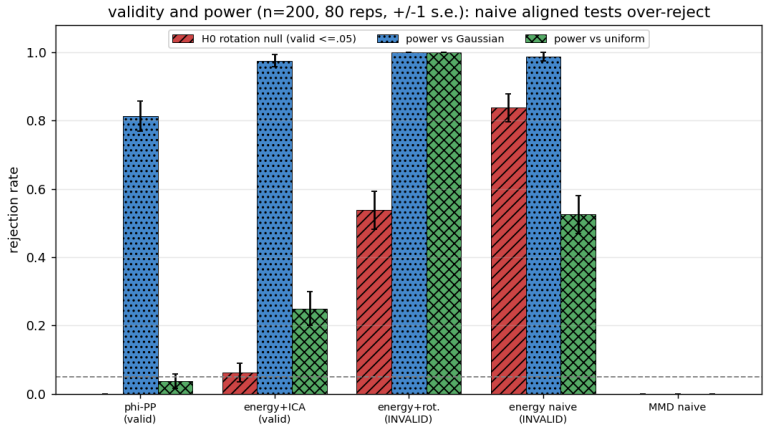


Figure 2. Validity and power ($n = 200$, 80 replications, ± 1 s.e.). The two naive aligned procedures over-reject the rotation null and are invalid; the projected test and energy-after-ICA control the level. Among the valid tests, energy-after-ICA is at least as powerful.

method	H_0 (rotation)	power (Gaussian)	power (uniform)
projected test (ours)	$0.00 \pm .00$	$0.81 \pm .04$	$0.04 \pm .02$
energy after ICA	$0.06 \pm .03$	$0.98 \pm .02$	$0.25 \pm .05$
energy, rotation-aligned	$0.54 \pm .06$	$1.00 \pm .00$	$1.00 \pm .00$
energy, naive	$0.84 \pm .04$	$0.99 \pm .01$	$0.53 \pm .06$
MMD, naive	$0.00 \pm .00$	$0.00 \pm .00$	$0.00 \pm .00$

Table 1. Rejection rates, isotropic regime, $n = 200$. The rotation-aligned and naive energy tests are invalid; the "power" columns are meaningful only for the level-controlling rows.

tests, energy-after-ICA is competitive and, on these alternatives, more powerful than the projected test—unsurprising, since it retains the full distribution after alignment while the projected test compresses to a 1-D profile. We therefore do not claim power superiority for the projected test. Its case rests on Sections 3–4: an explicit limit law and a bootstrap with a proven guarantee, where energy-after-ICA relies on a canonicalisation heuristic tied to ICA identifiability.

6. A Hermite view of contrast choice

Expand ξ in probabilist Hermite polynomials, $\xi = \sum_j c_j \mathrm{He}_j$, $c_j = \mathbb{E}_{N(0,1)}[\xi \mathrm{He}_j]/j!$. For whitened Z, $\mathbb{E}[\mathrm{He}_1(a^\top Z)] = 0$ (centred) and $\mathbb{E}[\mathrm{He}_2(a^\top Z)] = \mathrm{Var}(a^\top Z) - 1 = 0$ (identity covariance), so

$$k_P(\theta) = c_0(\xi) + \sum_{j \geq 3} c_j(\xi)\, m_j^P(\theta), \qquad m_j^P(\theta) = \mathbb{E}[\mathrm{He}_j(a_\theta^\top Z)] : \tag{2}$$

whitening annihilates Hermite orders $0, 1, 2$, and the test sees only the supra-quadratic part of ξ. Three consequences.

Non-degeneracy (Assumption 2) holds iff $\sum_{j \geq 3} c_j m_j^X$ is non-constant—the null must carry directional structure at some order $j \geq 3$ with $c_j \neq 0$. This is the surrogate gap

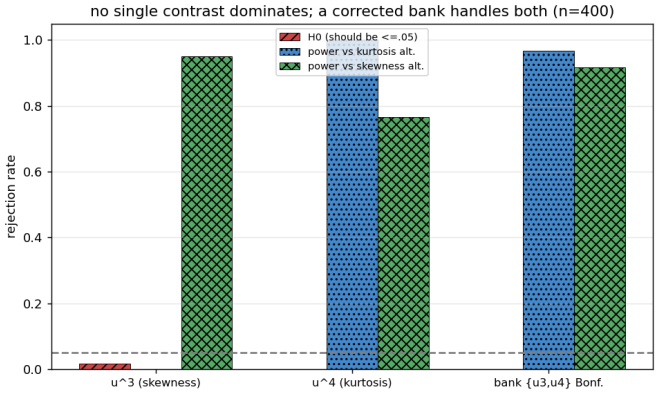


Figure 3. No single contrast dominates. u^4 misses the skewness alternative, u^3 the kurtosis one; the corrected bank $\{u^3, u^4\}$ handles both.

contrast	c_4	c_6	supra-quad. variance	data factor vs u^4
u^4	1.00	0	24.0	1×
u^6	15.0	1.0	6120	0.004×
$\lvert u\rvert$	−0.033	0.003	0.042	574×
$\log\cosh$	−0.015	0.001	0.006	3793×

Table 2. Post-whitening Hermite content. All contrasts are consistent; the data factor is the relative sample size for comparable signal. Efficiency is not monotone in $|c_j|$: the bounded $|u|$ has tiny coefficients but tiny noise.

of Remark 1 in coordinates, and links to ICA identifiability [6].

No single contrast dominates. A contrast is blind to any order with $c_j = 0$: u^4 (mass at j=4) is blind to skewness, an odd contrast to symmetric kurtosis. Figure 3 and Table 3 show the dissociation: u^3 detects a skewed alternative but not a Gaussian one, u^4 the reverse. All contrasts with supra-quadratic support are consistent (Table 2), differing only in efficiency; $\log\cosh$, standard in ICA, is the least efficient, its Hermite mass sitting at the annihilated order two.

Combine with a corrected bank. Since no fixed contrast is uniformly best, combine a few. Pooling the discrepancies into one statistic is delicate—the sum is dominated by the noisiest order and depends on scaling—so we use a scale-free multiplicity-corrected bank: reject if $\min_j p_j < \alpha/|\text{bank}|$. With $\{u^3, u^4\}$ this dominates either contrast alone (Table 3); the Bonferroni correction is conservative because the p_j are positively correlated, and a jointly calibrated combination would recover a little power.

7. A real-data illustration

The regime this paper targets—whitened, isotropic, non-Gaussian—occurs in real bivariate data, and the calibration caution of Section 5 is not an artifact of Gaussian simulation. We illustrate on the Wisconsin diagnostic breast-cancer data, taking two skewed real features (mean area and mean concavity; sample skewness 1.6 and 1.4).

Calibration on real marginals. To obtain a rotation null that is true by construction while keeping genuinely real marginals, we split the sample into two halves and apply

contrast/method	H_0	power (kurtosis alt.)	power (skewness alt.)
u^3 (skewness)	0.02	0.00	0.95
u^4 (kurtosis)	0.00	1.00	0.77
bank $\{u^3, u^4\}$ (Bonf.)	0.00	0.97	0.92

Table 3. Contrast choice, $n = 400$. Single contrasts dissociate; the bank handles both.

an independent random rotation to each; the two halves are then equal up to a rotation. Over 100 such splits, Table 4 reports the rejection rate at level 0.05. The naive energy test rejects the true rotation in 76%–83% of splits—the over-rejection of Section 5, now on real data. The projected test controls the level (no rejections). Strikingly, the ICA-canonicalized energy test—valid, and often more powerful, in the simulations of Section 5—over-rejects here (33%–64%): on these marginals the two whitened components carry similar higher-order structure, so the ICA canonicalization resolves the two halves to different orientations and the comparison ceases to be level-valid. The rotation-aligned energy test, by contrast, controls the level on these data (0.00–0.04), so its over-rejection in the isotropic simulation of Section 5 is regime-dependent; the naive energy test, however, over-rejects in both settings, and the projected test controls the level in both. The projected test's calibration does not depend on an ICA fit converging, and this is where that guarantee earns its place.

rejection under the rotation null ($\alpha = 0.05$)	area / concavity	radius / worst-conc.
projected test (ours)	**0.00**	**0.00**
energy after ICA	0.33	0.64
energy, rotation-aligned	0.00	0.04
energy, naive	0.76	0.83

Table 4. Calibration on real marginals (breast-cancer), two feature pairs. The projected test controls the level; the naive energy test grossly over-rejects; and the ICA-canonicalized energy test, valid in simulation, loses level control on real data.

A two-sample comparison. Testing the benign against the malignant class on the same two features, the projected test does not reject equality up to a rotation ($p = 0.45$), while the naive and ICA tests return $p = 0.17$ and $p = 0.03$. Given the calibration above—where the naive and ICA tests reject a *true* null at high rates—their p-values here cannot be read as evidence; only the level-valid projected test supports an interpretable conclusion. The lesson mirrors the paper's thesis: on real data the projected test is the one procedure that stays valid, so its verdict is the one a practitioner can trust.

8. Discussion

The practical message is twofold: testing equality up to a rotation demands calibration designed for the nuisance—naive aligned two-sample tests are anticonservative—and, once validity is secured, an ICA-canonicalized energy test is a strong baseline that the projected test does not beat on power. The projected test's value is its explicit theory

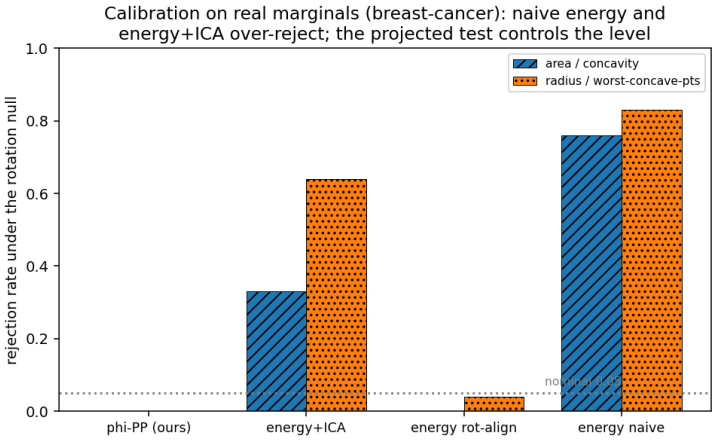


Figure 4. Rejection rate under a true rotation null on real marginals. Only the projected test controls the level across both feature pairs; naive energy and energy-after-ICA over-reject.

and guaranteed calibration, together with the Hermite reading of contrast choice.

The honest limits are several. The theory is for $d = 2$; general d replaces the scalar shift by an arg min over the $\mathcal{O}(d)$ manifold with $d(d-1)/2$ nuisance directions and can have non-isolated minima when components share higher moments, so Theorem 1 does not extend without further work; the same degeneracy makes Assumption 2 fail, and, tellingly, it is exactly when the alignment is ambiguous that ICA canonicalisation also fails, so no method gains there. The method targets a profile surrogate (Remark 1), mitigated but not removed by the bank. It suits the isotropic regime; under anisotropic covariance whitening already resolves the rotation. The kurtosis contrast needs eighth moments, so heavy-tailed sources require large samples. A robust divergence analogue for side-channel analysis is a natural applied direction requiring validation on real data, left to future work.

Disclosure statement

The author reports no competing interests.

Funding

No funding was received.

Data availability statement

The real-data illustration uses the Wisconsin Diagnostic Breast Cancer dataset, a public benchmark available in the `scikit-learn` library and originally from the UCI Machine Learning Repository. No new data were generated. All results—the simulation studies and the real-data analysis—are fully reproducible from the code provided as supplementary material and archived at `https://doi.org/10.5281/zenodo.23023835`, which runs with a fixed random seed.

Appendix A. Reminders and proofs

Donsker/Hermite. $\{\theta \mapsto \xi(a_\theta^\top z)\}$ is Donsker under a square-integrable envelope ($\|z\|^4$ for u^4, needing $\mathbb{E}\|Z\|^8 < \infty$; $O(\|z\|)$ for $\log\cosh$). The probabilist polynomials satisfy $\mathbb{E}_{N(0,1)}[\mathrm{He}_j \mathrm{He}_k] = j!\delta_{jk}$, giving (2); orders $0, 1, 2$ integrate to constants under centring and identity covariance.

Theorem 1. Let $D_n(s) = \frac{1}{M}\sum_g (\hat{k}_Y(\theta_g) - \hat{k}_X(\theta_g + s))^2$. At s_0, $nD_n(s_0) \rightsquigarrow \|\Delta\|^2$ by Lemma 1. By Assumption 2 and the argmax theorem $\hat{s} - s_0 = O_{\mathbb{P}}(n^{-1/2})$; a second-order expansion with $D_n'(s_0) = O_{\mathbb{P}}(n^{-1/2})$, $D_n''(s_0) \to D''(s_0)$ gives $n \min_s D_n(s) = nD_n(s_0) - \frac{1}{2}\{\sqrt{n}D_n'(s_0)\}^2 / D_n''(s_0) + o_{\mathbb{P}}(1)$. Since $\sqrt{n}D_n'(s_0) \rightsquigarrow -2\langle \Delta, k_X'(\cdot + s_0)\rangle$ and $D_n''(s_0) \to 2\|k_X'(\cdot + s_0)\|^2$, continuous mapping yields $L_\xi^{(M)}$; the reflection is a continuous minimum over two branches.

Propositions 1–2. With $W(\Sigma)$ smooth, a first-order expansion gives the augmented influence function $\bar{\psi}_a^X = \psi_a^X + \Lambda_a^\top \mathrm{vech}(XX^\top - \Sigma)$. Conditionally, $\mathbb{W}_\bullet^*$ sums independent mean-zero terms with conditional covariance $\to \Gamma_\xi^\bullet$, so by the multiplier CLT $\mathbb{W}_\bullet^* \rightsquigarrow \mathbb{W}_\bullet$ conditionally in probability; the norm and score projection are continuous and $\hat{s} \to s_0$, giving $T_n^* \rightsquigarrow L_\xi^{(M)}$. The anticonservatism of the rotation-aligned permutation test follows because the observed statistic minimises over a continuum of rotations while each permutation replicate minimises over the same continuum on exchangeable data.

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